

JavaScript Array Methods - Complete Guide

The Big Three: map, filter, reduce

1. MAP - Transform each element

```
const numbers = [1, 2, 3, 4, 5];
const doubled = numbers.map(n => n * 2);
// doubled = [2, 4, 6, 8, 10]

// With objects:
const students = [
  { name: 'Ahmed', marks: 85 },
  { name: 'Umer', marks: 92 }
];
const names = students.map(s => s.name);
// names = ['Ahmed', 'Umer']
```

2. FILTER - Select specific elements

```
const marks = [50, 30, 80, 90, 20];
const passed = marks.filter(m => m >= 50);
// passed = [50, 80, 90]

// Get only active students:
const active = students.filter(s => s.active);
```

3. REDUCE - Combine into single value

```
const sum = marks.reduce((acc, m) => acc + m, 0);
// sum = 270

const total = students.reduce((sum, s) => sum + s.marks, 0);
// total = 177
```

Other Useful Methods:

forEach - Loop through array:

```
marks.forEach(m => console.log(m));
```

find - Get first matching element:

```
const first = marks.find(m => m > 80); // 80
```

some - Check if ANY match:

```
const hasPass = marks.some(m => m >= 50); // true
```

every - Check if ALL match:

```
const allPass = marks.every(m => m >= 50); // false
```

Combining Methods (Most Powerful!):

```
// Get names of students with marks > 80
```

```
const topStudents = students
```

```
.filter(s => s.marks > 80)
```

```
.map(s => s.name);
```